Stochastic Safety Margins: Quantifying the Penalty of Behavioural Noise in Stadium Egress

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Abstract

Current industry standards for stadium egress safety rely on deterministic mean-flow models that overlook the individual behavioural noise inherent in human movement. This research implements an Active-Langevin framework in Python, parameterising navigational uncertainty via a diffusion coefficient (D) to evaluate its direct impact on egress performance. Simulations of 50 agents in a 2.4 m stadium vomitory identify a non-linear "Stochastic Penalty"; at moderate noise ($D = 0.5$), egress efficiency dropped by 11% relative to the deterministic baseline. Localised analysis revealed that stochastic triggers within the bottleneck simultaneously intensify peak occupancy and extend the high-pressure state by 9 seconds – a 33% increase in duration. Physically, these clogs are driven by individual hesitations that propagate as ripples of deceleration to form stable structural arches. This study proves that safety is a function of both geometry and behavioural state, offering a stochastic lens for more resilient engineering safety margins.

The full Python source code and simulation environment used in this study are hosted in a public GitHub repository at: https://github.com/naveen-k-m-rg/Stochastic-Safety-Margins-Quantifying-the-Penalty-of-Behavioural-Noise-in-Stadium-Egress

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Core Langevin Variables

Symbol	Definition	Units
m	Pedestrian mass	kg
t	Time	s
v_i	Instantaneous velocity vector of agent i	m/s
$\frac{dv_i}{dt}$	Acceleration of agent i	m/s²
x_i	Spatial coordinates of agent i	m
F_{det}	Resultant sum of deterministic forces	N
$\xi_i(t)$	Stochastic behavioural noise term	N
D	Diffusion coefficient (behavioural noise intensity)	-
$\eta_i(t)$	Vector of Gaussian random fluctuations	-
W_i	Wiener process (random walk) increment	-
Δt	Simulation time step	s

Deterministic Force Components (F_{det})

Symbol	Definition	Units
$F_{i,drive}$	Driving force toward the exit goal	N
$F_{i,soc}$	Pairwise social repulsion force	N
$F_{i,bound}$	Boundary (wall) repulsion force	N
v_0	Desired walking velocity	m/s
τ	Relaxation time (reaction delay)	s
$\hat{e}_i$	Desired velocity unit vector	-

Social and Boundary Interaction Parameters

Symbol	Definition	Units
w_{ij}	Anisotropy weight (directional influence)	-
λ	Weighting constant for rear-facing vision cone	-
$cos\phi_{ij}$	Dot product between velocity and neighbour direction	-
k_{soc}	Social repulsion stiffness	N/m
$d_{buffer\ soc}$	Personal space (social) radius	m
r_{ij}	Straight-line distance between agent i and j	m
$\hat{n}_{ij}$	Unit repulsion vector	-
c_{soc}	Social damping coefficient	Ns/m
k_{wall}	Wall repulsion stiffness	N/m
$d_{buffer\ wall}$	Wall activation radius	m
δ	Signed distance to boundary	m
c_{wall}	Wall damping coefficient	Ns/m
$\hat{n}$	Wall normal vector	-

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Glossary of Key Terms

Active-Langevin Framework: A specialised modelling approach that treats pedestrians as self-propelled "active" particles governed by a stochastic differential equation. It mathematically splits forces into a deterministic "drift" component (the agent's goal-seeking intent) and a stochastic "noise" component (individual-level random fluctuations).

Butterfly Effect: A phenomenon within high-density crowds where a single individual perturbation (such as a random hesitation) triggers a non-linear ripple of deceleration through the queue, leading to collective flow failure.

Danger Zone: The critical 1.0m approach area immediately preceding a bottleneck (vomitory throat) characterised by the highest lateral pressures and probability of arching.

Deterministic Fallacy: The flawed engineering assumption that pedestrian flow is a perfectly predictable, linear function of corridor width, neglecting individual behavioural variance.

Diffusion Coefficient (D): The volume knob within the Langevin framework that parameterises the intensity of an agent's internal behavioural noise or agitation.

Euler-Maruyama (EM) Method: The numerical solver used to discretise continuous stochastic differential equations into finite time steps (Δt). It is the stochastic counterpart to the standard Euler method and ensures the statistical integrity of the behavioural noise term remains physically grounded.

Pulse-Loading: A stewardship or design strategy where crowd entry is staggered at discrete intervals to synchronise arrival rates with the physical capacity of the infrastructure.

Social Force Model (SFM): The foundational deterministic framework used to model pedestrian dynamics through social motivations (acceleration toward a goal) and social aversions (repulsion from obstacles and boundaries).

Stochastic Penalty: The quantifiable reduction in global egress efficiency – identified as an 11% loss in this study – caused by individual-level navigational uncertainty.

Thinking Fluid: A hypothesis treating high-density crowds as a continuum where individuals possess internal motivation and navigate toward a goal while avoiding density.

1. Introduction

1.1 The Global Challenge of Mass-Motion

As global urbanisation accelerates toward the late 2020s, the management of high-density pedestrian flow has become a primary mandate for civil engineering. With the expansion of transport hubs and the construction of mega-arenas for global events, the responsibility of the engineer has shifted from the purely structural to the dynamic. Public safety in these environments is no longer solely a function of how a building stands, but how a crowd moves within it. In high-occupancy scenarios, the crowd fluid is a volatile entity; localised disruptions can rapidly escalate into systemic safety risks.

1.2 The Paradox of the Modern Stadium

Sports stadiums represent a unique architectural paradox. While their primary purpose is to host a stationary audience, their egress (exit) routes must facilitate a sudden, massive, and highly synchronised transition from static occupancy to active mobility the moment an event concludes. Historically, the failure of architectural design accounting for this transition has led to significant disasters, highlighting the need for resilient egress design. These events are rarely the result of structural failure, but rather a mechanical breakdown of the crowd itself, manifesting in two high-risk states:

- **Physical Clogging:** Where the geometry of an exit, such as a narrow vomitory, cannot process the density of individuals, leading to arches (Figure 1.2) that stop all forward progress (Zhang et al., 2007).
- **The Stranded-Crowd Effect:** Where individuals at the front of a bottleneck are rendered immobile while the momentum from the trailing crowd continues to build (Huang et al., 2013).

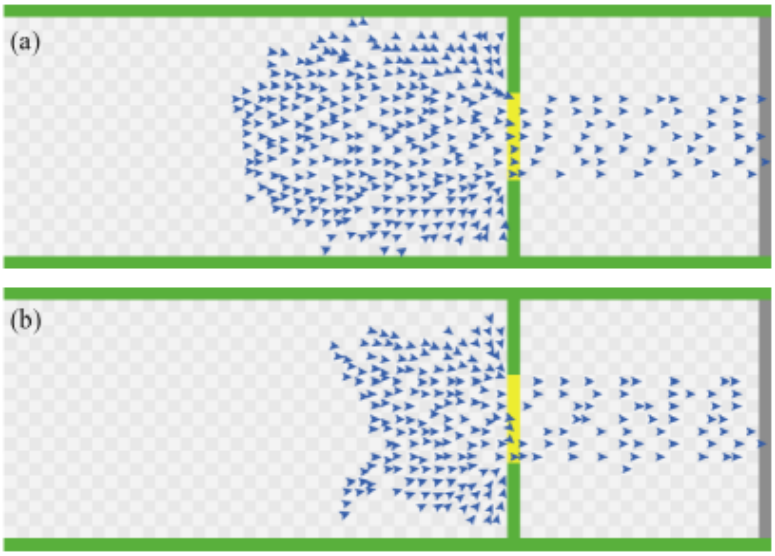


Figure 1.2: Profile of the crowd before the exit door: *(a) Onset of arching where the minor and major axes are visible; and (b) Occurrence of double arches toward the end of the simulation.*

1.3 Theoretical Context: The "Thinking Fluid"

To predict these risks, civil engineers have historically looked to the laws of classical physics, treating high-density crowds as a macroscopic continuous medium analogous to a fluid. By applying principles of fluid mechanics, movement can be modelled as a flow responding to pressure and density gradients.

This fluidity is not merely theoretical; it is a visible reality in extreme crowd scenarios. As illustrated in Figure 1.3, the opening ceremony of the San Fermín festival demonstrates how thousands of individuals can enter a collective state, where the group exhibits fluid-like undulations and stop-and-go waves that transcend individual choice (Miller, 2025).

A significant evolution in this field was the "Thinking Fluid" hypothesis pioneered by Hughes (2003). Unlike a classical fluid, a crowd possesses internal motivation; individuals navigate according to a potential field to reach a goal. By treating the crowd as a continuum with a shared intent, this approach provides a robust macroscopic baseline for designing corridors and exits.

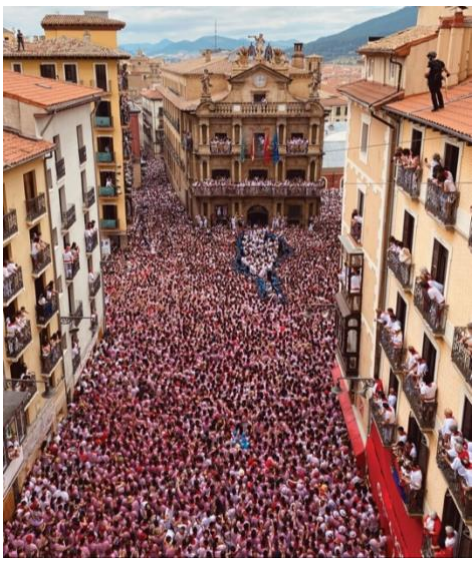


Figure 1.3: Collective dynamics at the San Fermín opening ceremony, Pamplona*. As crowd density exceeds 4 persons/m², the crowd ceases to behave as a collection of individuals and begins to exhibit quasiperiodic orbital undulations and stop-and-go waves. This real-world scenario serves as a primary example of a Thinking Fluid, where macroscopic flow is governed by collective physical laws rather than individual intent. (Photo: Bartolo Lab / Physics Today, 2025).*

1.4 The Problem: The "Deterministic Fallacy"

Despite the elegance of fluid analogies, current industry benchmarks, most notably The Green Guide (Sports Grounds Safety Authority, 2018), rely on the "Deterministic Fallacy": the assumption that crowd flow is a linear, predictable function of geometry that ignores human variance. These standards assume a linear relationship between corridor width and flow rate, effectively treating pedestrians as robotic agents that react identically to a given set of conditions.

However, actual human behaviour is characterised by a high degree of variance; random hesitations and navigational uncertainty. Traditional mean-flow models average out these individual fluctuations, failing to account for the Stochastic noise that actually triggers the transition from flow to failure. To bridge this gap, a framework is required that treats individual randomness not as an outlier, but as a primary mathematical variable.

1.5 Research Aims

The overarching aim of this research is to bridge the gap between theoretical physics and civil engineering by constructing a functional computational engine that simulates the complexities of human crowd movement. Specifically, the project seeks to translate the abstract principles of individual human decision-making into a stable digital environment, providing a new stochastic lens to quantify how individual-level uncertainty triggers systemic failures during stadium egress. In doing so, the study intends to demonstrate that behavioural randomness can be successfully parameterised to provide more resilient safety margins than traditional mean-flow models.

2. Literature Review

2.1 The Evolution of Crowd Modelling

The academic effort to simulate pedestrian movement has historically oscillated between two distinct scales of representation: the macroscopic (continuum) and the microscopic (discrete). This section evaluates the foundational theories of global flow and identifies the deterministic gaps that necessitate a more robust, stochastic approach to stadium safety.

2.1.1 The Continuum Foundation: The "Thinking Fluid"

Building upon the fluid analogy introduced in Section 1.3, the Thinking Fluid hypothesis (Hughes, 2003) is governed by a set of partial differential equations (PDEs) where individuals navigate based on a potential field. This field represents the desire to minimise travel time to a goal while avoiding areas of high density.

By treating the crowd as a shared intent, the model provided a robust macroscopic baseline for engineering design, allowing for the efficient prediction of where shocks or global bottlenecks are likely to form.

However, while mathematically elegant, these models are inherently deterministic and assume that the fluid will always react identically under a given set of conditions.

2.1.2 Macroscopic Observations of Phase Transitions

Recent research has sought to validate these fluid analogies using empirical data from extreme crowd events. Helbing et al. (2007) analysed video data from real-world disasters (such as the 2006 Hajj disaster), to identify a mathematical transition from laminar flow to crowd turbulence. This work provided a pivotal mathematical foundation for observing real-world stop-and-go waves, proving that crowds exhibit phase transitions similar to physical fluids.

As illustrated in Figure 2.1.2, these phase transitions manifest as a collapse in steady-state velocity, where the flow rate becomes dictated by the formation and dissolution of localised clusters. While these models successfully identify the macroscopic shift from laminar to turbulent flow, they are often limited by a lack of microscopic stochasticity.

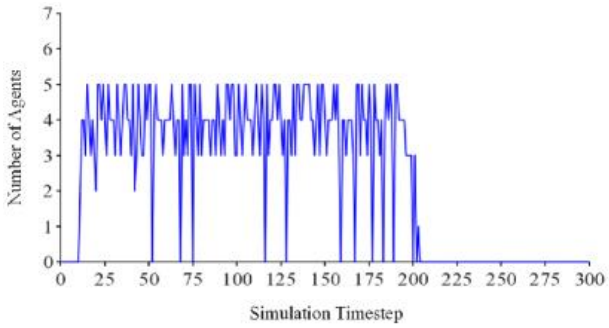


Figure 2.1.2: Simulation data illustrating the transition from laminar flow to stop-and-go waves. *The plot tracks agent velocity over time within a high-density bottleneck, showing the emergence of periodic oscillations and localised stalling as crowd density increases (Castro and Pabico, 2015).*

Modern refinements by researchers such as Gu et al. (2025) have continued to model these collective phenomena, including self-organised oscillations in dense crowds. However, while these macro-models are increasingly sophisticated at describing the results of crowd turbulence (the chaotic breakdown of laminar flow into erratic oscillations), they remain largely descriptive and still suffer from the same fundamental deterministic gap as the preceding research. By treating the crowd as a continuum, these models rely on mean-flow assumptions where agents act according to a rational average. Consequently, they fail to account for individual-level random fluctuations, leaving a distinct research void regarding the potential for such randomness to act as the primary trigger for the transition from flow to systemic failure.

2.1.3 The Industry Standard: The "Deterministic Fallacy"

The codification of this deterministic perspective within industry benchmarks like The Green Guide (Sports Grounds Safety Authority, 2018) relies on the mean-flow approach described in Section 1.4. Within these standards, the safety and capacity of an exit are treated as a linear function of corridor width, assuming that flow remains predictable as long as geometric requirements are met.

The critique of this standard lies in its reliance on static, deterministic baselines. Current design practices assume that if a corridor meets a specific width requirement, the flow will remain predictable. However, as the literature in Section 2.2 will demonstrate, human behaviour is characterised by high variance that a mean-flow calculation cannot capture. To bridge this gap, a model is required that treats behavioural noise not as an outlier, but as a primary mathematical variable.

2.2 Stadium Egress: A Human Variable

While the foundational macroscopic models discussed in Section 2.1 provide the mathematical scaffolding for crowd movement, they often operate under the assumption of an "ideal" pedestrian. This section evaluates the microscopic transition toward agent-based modelling and identifies the specific physical, behavioural, and cognitive variables that disrupt deterministic flow in stadium environments.

2.2.1 The Microscopic Pivot: Social Forces and Coordination

A fundamental shift toward the individual occurred with the development of the Social Force Model (SFM) (Helbing & Molnár, 1995). This approach treats pedestrians as discrete entities subject to "social forces"; mathematical measures of internal motivations, including acceleration toward a goal and repulsion from

obstacles. Crucially, these equations of motion function as non-linearly coupled Langevin equations (Langevin, 1908); stochastic differential equations that model the interplay between deterministic drift (intent) and random fluctuations (noise). While the rigorous derivation and implementation of these mechanics are detailed in Section 2.3 and Section 3.1, their introduction here marks the point where pedestrian dynamics officially transitioned from simple kinematics to a branch of statistical physics.

However, much of the subsequent application of the SFM in engineering has focused primarily on the deterministic force terms (F_{det}), frequently treating the stochastic term (ξ) as a secondary fluctuation or neglecting it entirely. Modern refinements, such as the velocity-space avoidance of Xu et al. (2014) or the coordination models of Karamouzas et al. (2017), have significantly improved the replication of smooth, anticipatory avoidance. Yet, a rigorous assessment of these frameworks reveals a persistent assumption of near-perfect or rational coordination. While effective for simulating smooth navigation, such models often lack the mathematical mechanism to parameterise the individual-level behavioural noise or human error that this study investigates as a potential trigger for systemic failure in high-density stadium bottlenecks.

2.2.2 Defining the Physical Failure: Clogging and Stranding

The mechanical failure states identified in Section 1.2 (clogging and stranding) are further quantified by Zhang et al. (2007) and Huang et al. (2013) through the Stranded-Crowd Model (SCM). This research confirms that high-density egress is governed by physical contact forces that mirror the mechanics of clogging in granular materials, leading to the formation of the semi-circular arches observed at stadium bottlenecks.

A substantive engagement with this literature reveals a persistent procedural gap. While both studies successfully validate the outcome of clogging through empirical data (such as the 1996 Guatemala City disaster), their models remain largely deterministic. They utilise empirical relations to describe the crowd state after a density threshold is breached, missing the stochastic trigger – the individual hesitation or random deviation – that initiates the transition from flow to a stable, dangerous arch. Furthermore, current research focuses heavily on the occurrence of a clog, leaving a distinct gap regarding the temporal stability of these formations and how individual-level variance influences the duration of high-pressure states at the bottleneck.

2.2.3 Sources of Behavioural Variance ("Noise")

Contemporary research proves that the average pedestrian assumed in static design guides is a myth. Larsson et al. (2021) found that actual flow rates in stadiums are significantly lower than benchmarks like The Green Guide, largely due to demographic diversity. Furthermore, variables such as motivation levels (Young et al., 2021) and alcohol consumption (Gales et al., 2025) have been shown to fundamentally increase the standard deviation ("noise") of pedestrian speeds.

This is quantitatively supported by Campanella, Hoogendoorn, and Daamen (2009), who demonstrate that heterogeneity in characteristics such as desired speed, body size, and reaction time significantly influences breakdown probabilities. Their findings suggest that diversity congests, with speed variance potentially reducing average throughput by up to 50% in high-inflow scenarios. Crucially, they observe that even a small fraction of a population exhibiting erratic navigation (characterised by high reaction times) can cause collective efficiency to drop.

The critical gap in current engineering practice lies in the application of these findings. While researchers have successfully identified these sources of variance, industry standards continue to design for the mean flow. Current models frequently smooth over inter-pedestrian differences, failing to account for the erratic triggers and speed variabilities that initiate systemic failure. Without a primary role for stochasticity to parameterise these fluctuations, engineers cannot accurately quantify the probability of the unpredictable triggers that lead to systemic failure in stadium environments. To bridge this gap, a framework is required that can treat behavioural noise not as an outlier, but as a primary mathematical variable.

2.3 The Active-Langevin Perspective

2.3.1 Origins and the Theoretical Framework

To fulfil the research aim of bridging the gap between theoretical physics and civil engineering (Section 1.5), this study appraises the Langevin equation as the primary tool for modelling systems governed by both "drift" (intent) and "noise" (randomness).

Pioneered by Langevin (1908) to describe Brownian motion, this framework splits the governing forces into two distinct components:

- **The Deterministic Component (Drift)**: Represents the macroscopic intent or intended trajectory of the agent, such as the desire to reach an exit.
- **The Stochastic Component (Noise)**: Represents individual-level random fluctuations, effectively modelling the navigational uncertainty and agitation of a pedestrian.

By splitting the forces in this manner, the framework provides a rigorous precedent for a Thinking Fluid, allowing a modeler to define a pedestrian's intended path (drift) while acknowledging they are subject to an erratic, internal random force (behavioural noise).

Modern refinements, as detailed by Coffey et al. (1996), have evolved this into a mean-reverting process, which effectively represents a pedestrian's desired velocity as a stable state that is realistically disrupted by stochastic fluctuations. This is further justified through the lens of Active Brownian Particles (Romanczuk et al., 2012), where self-propelled agents exhibit internal, non-linear active noise rather than reacting purely to external pressure. This transition enables the simulation of individual behavioural triggers (the random hesitations) that initiate the systemic jamming phase transitions identified in Section 2.2.2.

2.3.2 Validation in Pedestrian Dynamics

The modern appraisal of crowd dynamics has successfully mapped human behaviour to these stochastic equations. Corbetta et al. (2018) utilised millions of real-world trajectories (harvested via high-resolution overhead depth sensors in a railway station corridor) to validate that coupled Langevin equations can quantitatively reproduce the statistical features of human walking and avoidance dynamics. Their work proves that the inherent fluctuations in human paths are not merely errors in a deterministic system but are fundamental statistical properties. Building upon this, Behera et al. (2020) formalised the Active-Langevin model as a specialised tool for understanding dense crowd movements by treating the crowd through the lens of active matter physics.

While these advancements prove the model's descriptive power, a significant research opportunity remains. Much of the existing literature, including the perception-based noise models of Iyer et al. (2024), focuses on general avoidance or open-space navigation. There is a distinct need to take this validated Active-Langevin framework and apply it to the high-stakes, constrained geometry of stadium bottlenecks. This study seeks to bridge that gap, utilising the Langevin equation to quantify the "Stochastic Penalty" (the efficiency loss triggered by navigational uncertainty) and identify the mechanical failure points – clogging and stranding – that traditional deterministic models fundamentally overlook, evaluating not only the reduction in global throughput but also the potential for prolonged occupancy within critical exit routes.

2.4 Research Questions

Following the critical appraisal of existing crowd dynamics and the identification of the Deterministic Fallacy in industry standards, this research seeks to answer the following questions:

- **Research Question 1:** Is it computationally feasible to implement a stable, agent-based framework that successfully integrates individual-level behavioural noise with the global navigational goals of a stadium crowd?
- **Research Question 2:** To what extent does the presence of this individual randomness degrade global egress efficiency and influence the transition to systemic flow failure compared to traditional deterministic expectations?

3. Methodology

This chapter details the transition of the Active-Langevin framework from theoretical concept to functional engineering tool. The methodology is structured to move from mathematical formulation (Section 3.1) and numerical discretisation (Section 3.2) toward the iterative development of the simulation environment and its specific research objectives (Section 3.3).

3.1 Model Formulation: The Active-Langevin Framework

The primary methodology of this research utilises a microscopic, agent-based framework where each pedestrian (i) is treated as a self-propelled active particle. Following the theoretical foundation established in Section 2.3.1, the motion of these particles is governed by a second-order Stochastic Differential Equation (SDE). This mathematical architecture serves as the blueprint for the project, designed to maintain the macroscopic, goal-oriented intent of a crowd (reaching the stadium exit) while accounting for the individual-level fluctuations that characterise human movement.

3.1.1 The Governing SDE Framework

In its most general form, the motion of a particle (or agent) is defined by the following SDE. For an individual pedestrian i with a mass m, the change in velocity v_i over time t:

$$m\frac{dv_i(t)}{dt} = F_{i,det}(x_i, v_i) + \xi_i(t) \tag{1}$$

Where:

- $m\frac{dv_i(t)}{dt}$ = Acceleration; tracks how pedestrian moves
- F_{det} = Sum of deterministic forces representing "drift" or intended trajectory of the agent
- $\xi_i(t)$ = Stochastic behavioural noise term representing individual-level random fluctuations

This mathematical architecture is designed to maintain the macroscopic, goal-oriented intent of a crowd while accounting for the individual-level fluctuations that characterise human movement.

3.1.2 Establishing the Deterministic Framework (F_{det})

Before selecting the governing terms for the deterministic drift component of the Langevin equation, several modelling scales were evaluated. While macroscopic continuum models (e.g., Hughes, 2003) provide a strong theoretical basis for global flow, they were discarded as a primary framework because they average out the discrete, agent-to-agent interactions necessary to model physical clogging. Similarly, purely granular models (Lin et al., 2007) were found insufficient as they prioritise physical contact over the internal motivational aspects of human navigation.

The Social Force Model (SFM) (Helbing & Molnár, 1995) was selected as the optimal deterministic foundation. The SFM allows for a Thinking Fluid approach where pedestrians are governed by social desires (reaching an exit) and social aversions (avoiding walls and collisions). This framework is uniquely compatible with the Langevin approach, as it allows the deterministic drift to be clearly separated from the stochastic noise.

The SFM is integrated into the Langevin framework as a summation of three sub-forces:

$$F_{i,det} = F_{i,drive} + F_{i,soc} + F_{i,bound} \tag{1.1}$$

Where:

- $F_{i,drive}$ = Driving Force
  - Represents the internal motivation toward the exit; the mathematical implementation of the Thinking Fluid philosophy established by Hughes (2003).
- $F_{i,soc}$ = Social Repulsion
  - Following the work of Corbetta et al. (2018), this term ensures agents maintain a personal space buffer, effectively modelling social coordination and collision avoidance.

- $F_{i,bound}$ = Boundary Repulsion
    - This accounts for the physical stadium geometry, specifically the exit vomitories and gates defined in the research of Zhang et al. (2007).

3.1.3 The Stochastic Noise Term ($\xi_i(t)$)

The behavioural noise $\xi_i(t)$ is a fundamental component of the Active-Langevin framework, functioning as the primary mechanism for simulating navigational uncertainty. It is modelled as Gaussian White Noise with a mean of zero and unit variance, adhering to the classical framework established by Langevin (1908). This stochastic component is represented mathematically as:

$$\xi_i(t) = \sqrt{2D}\eta_i(t) \tag{1.2}$$

Where:

- $\eta_i(t)$ = Vector of random fluctuations ($N(0,1)$)
- D = Diffusion coefficient

In this governing equation, D acts as a volume knob that dictates the intensity of behavioural noise. The use of the square root factor ($\sqrt{2D}$) is mathematically necessary to maintain dimensional consistency with the velocity update and to ensure that the noise power remains physically grounded during numerical integration. Notably, when $D = 0$, the stochastic term is nullified, and the agent's movement reverts entirely to the deterministic Social Force Model (SFM) baseline. Conversely, when D is elevated, agents exhibit an agitated state, where frequent and high-magnitude stochastic kicks trigger erratic path deviations that disrupt laminar flow.

While D represents the aggregate strength of the noise, the $\eta_i(t)$ term provides the instantaneous kick that physically alters the agent's path. This ensures that no two pedestrians follow perfectly synchronised trajectories, even when subjected to identical environmental forces. By introducing this individual-level heterogeneity, the model can simulate the random hesitations and deviations that serve as the catalysts for collective flow failure, which are quantitatively explored through the parameter sweep in Section 4.3.

3.2 Numerical Implementation: The Euler-Maruyama Method

Building upon the mathematical baseline, this section describes the transition from the theoretical Langevin equations to a functional computational model. To solve the continuous stochastic differential equations (SDEs) within a digital environment, the system must be discretised into finite time steps (Δt). This research utilises the Euler-Maruyama (EM) method, the standard numerical approach for integrating SDEs while maintaining the statistical integrity of the noise term, providing the numerical stability required to meet the project's tactical objectives.

3.2.1 Background to Euler-Maruyama

The Euler-Maruyama method serves as the stochastic counterpart to the standard Euler method used in deterministic physics. Following the mathematical framework detailed by Coffey, Kalmykov, and Waldron (1996), the EM method provides a robust way to update an agent's state by splitting the calculation:

- **The Euler Component (Drift):** This handles the deterministic forces (F_{det}), calculating where the agent intends to be based on driving, social, and boundary forces.
- **The Maruyama Component (Kick):** This handles the stochastic behavioural noise (ξ), introducing the random kick that accounts for navigational uncertainty

The selection of the Euler-Maruyama method for this stadium egress simulation is based on three primary engineering advantages:

1. **Computational Efficiency:** The EM method is computationally efficient and highly compatible with vectorised operations. By utilising a vectorised architecture, the simulation can update the velocity and position of all agents simultaneously. This is essential for maintaining numerical performance while modelling the high-occupancy, high-density environments characteristic of a stadium egress.

2. **Suitability for SDEs:** Unlike higher-order deterministic solvers that require smooth, differentiable functions, EM is specifically designed for equations involving non-differentiable noise terms characteristic of the Langevin approach.
3. **Statistical Integrity:** EM is mathematically proven to converge to the correct statistical distribution of paths. This is vital for crowd safety modelling, as it ensures that the resulting density fluctuations and "Total Egress Time" (TET) distributions are mathematically valid and not artifacts of the numerical solver.

3.2.2 Discretisation Logic

The discretisation of the continuous Langevin equation is a critical step in translating theoretical physics into a functional computational engine. Unlike deterministic integration, where variables scale linearly with time, the stochastic component of the Active-Langevin framework must account for the mathematical properties of random processes.

To maintain the statistical integrity of the behavioural noise, the velocity update for each agent at every time step (Δt) is calculated using the following discrete equation:

$$v_i(t + \Delta t) = v_i(t) + \frac{1}{m} F_{i,det} \Delta t + \sqrt{2D}\sqrt{\Delta t} W_i \quad (2)$$

Deterministic Update ($\frac{1}{m} F_{i,det} \Delta t$):

- This represents the intended movement of the agent by multiplying the sum of all social, driving, and boundary forces by the time step.
- It defines the trajectory the pedestrian would follow in a perfect world where navigation is entirely predictable and unaffected by behavioural noise.

Stochastic Update ($\sqrt{2D}\sqrt{\Delta t} W_i$):

- This introduces the behavioural kick by scaling a Wiener process (W_i) – which models the continuous accumulation of random fluctuations over the interval – by the square root of the time step.
- The use of $\sqrt{\Delta t}$ rather than Δt for the noise term is an essential requirement for stochastic stability. This is a fundamental property of a random walk, where the spatial dispersion of an agent scales with $\sqrt{\Delta t}$ rather than Δt. If the stochastic component were scaled linearly by Δt, the behavioral noise would effectively disappear as the time steps became smaller, leading to a loss of physical realism at higher simulation resolutions. By utilising the square root scaling, the model ensures that the intensity of the navigational uncertainty remains constant regardless of the simulation's temporal resolution.

This discretisation logic allows for a robust transition from theoretical SDEs to a functional, agent-based environment. It ensures that the emergent properties of the crowd – such as the formation of arches and the onset of clogging – can be observed and analysed in real-time with numerical consistency.

3.3 Computational Framework: Python Implementation

The following sections detail the development of the egress simulation, transitioning the theoretical Active-Langevin framework into a functional agent-based model. The implementation follows an iterative process to ensure that the deterministic intent, environmental constraints, and stochastic variables established in previous sections are correctly integrated into a robust multi-agent system.

The primary objectives governing this development of this computational framework were:

1. **Numerical and Physical Stability:** To design and calibrate a simulation engine that correctly interprets the Active-Langevin governing equations. This involves establishing a robust computational architecture capable of maintaining mathematical stability and convergence during high-density interactions. The goal is to ensure the macroscopic intent of the crowd is preserved even under high-noise conditions that challenge traditional deterministic assumptions.

2. **Behavioural Parameterisation and Bottleneck Integration:** To establish an agent-based environment where the Diffusion Coefficient (D) can be parameterised as a primary independent variable. This objective focus involves integrating stochastic noise within constrained stadium geometries (specifically a 2.4m vomitory) to isolate the specific triggers that initiate physical clogging and evaluate the structural stability of the resulting flow-locks.
3. **Automated Data Extraction:** To implement a high-fidelity diagnostic system for tracking Total Egress Time (TET) alongside localised bottleneck metrics, such as instantaneous agent velocities and approach-zone occupancy duration. This provides the mechanism for harvesting the quantitative data necessary to evaluate stochastic failure against traditional, deterministic safety benchmarks, accounting for both the intensity and temporal duration of bottleneck congestion.

Each stage of the development represents a targeted increase in model complexity, moving from single-agent verification to a fully parameterised environment capable of identifying systemic failures.

3.3.1 Modular Code Architecture

To ensure the model remained adaptable and numerically stable as complexity increased, the simulation script was organised into five distinct Components. This structure was maintained across every iteration of the research to ensure consistency:

Component 1: Simulation Constants and Parameters – Defining the physical and behavioural environment.
Component 2: Agent Initialisation – Setting starting coordinates and velocity arrays.
Component 3: Simulation Loop – The primary "engine" where the governing physics are calculated frame-by-frame.
Component 4: Metric Calculation – Extracting quantitative data (e.g., exit times) after the simulation concludes.
Component 5: Visualisation – Producing real-time plots to observe emergent behaviours.

The 5-Step Simulation Loop (Euler-Maruyama Implementation)

The core of the framework is Component 3, where a nested five-step cycle executes for every agent at every time step (Δt). This sequence is the direct computational implementation of the discretisation logic defined in Section 3.2.2. By strictly following this order, the model prevents feedback errors where an agent's position is updated before all forces (including the stochastic kick) have been accounted for.

Step 1: Force Summation (F_{det}) – The loop initiates by calculating the resultant deterministic vector. While the summation is comprised solely of the driving force (F_{drive}) in the initial framework, the architecture is designed for modularity, allowing for the subsequent integration of architectural constraints (F_{bound}) and social interactions (F_{soc}) in Section 3.3.2. This step evaluates the agent's current coordinates relative to the target to determine the specific acceleration required to reach the desired velocity (v_0).

Step 2: Langevin Noise (ξ) – The stochastic term is generated as a random vector independently of the deterministic forces. Using a Gaussian normal distribution, the framework produces a navigational kick that is scaled by $\sqrt{2D}\sqrt{\Delta t}$ to ensure that the behavioural agitation remains physically consistent and independent of the simulation's temporal resolution.

Step 3: Velocity Update – The core decision step of the engine. The code combines the deterministic acceleration from Step 1 and the stochastic kick from Step 2 to calculate the new velocity vector. By updating velocity before position, the model maintains a First-Order Integration scheme, which is essential for the numerical stability of the Langevin solver.

Step 4: Position Update – The new velocity is multiplied by the time step (Δt) and added to the agent's current coordinates. This step translates the abstract mathematical forces into physical displacement within the virtual stadium, representing the movement of the Thinking Fluid through the corridor toward the designated exit.

Step 5: Logging – The updated position and velocity array data for every agent is appended to a global history list. By storing the entire trajectory in memory, the model enables the complex post-simulation analysis described in Component 4, allowing the researcher to identify the specific onset of emergent behaviours or flow interruptions without re-computing the expensive physics engine.

3.3.2 Environmental Geometry and Multi-Agent Interaction

Following the establishment of the baseline navigational engine, this phase integrates environmental and social constraints into the governing Active-Langevin equation. This transition shifts the model from intent-only logic to a constrained environment that accounts for the physical and social limitations inherent in stadium egress.

Note on Architectural Design: The mathematical structures and logic layers defined in this section – including the force hierarchies, numerical caps and initialisation strategies – were established through iterative experimental testing. While this section defines the technical blueprint of the simulation, the experimental determination of specific constants (k, c, τ) and the calibration of stability regimes are detailed as primary findings in Section 4.2.

The Deterministic Force Expansion

The deterministic force summation (F_{det}) is expanded to represent a Thinking Fluid navigating a physical bottleneck. To maintain numerical stability during high-density interactions, a Linear Spring-Damper model was selected for all repulsive forces. Fundamentally, these equations translate the Thinking Fluid philosophy into three simple instructions for the agent: move toward the goal ($F_{i,drive}$), maintain personal space ($F_{i,soc}$), and avoid the stadium walls ($F_{i,bound}$).

Driving Force ($F_{i,drive}$)

This term represents the goal-seeking intent of the agent. To simulate a realistic transition from a stadium bowl into a narrow exit, a zone-based target logic is implemented:

$$F_{i,drive} = \frac{m}{\tau}(v_0 \hat{e}_i - v_i) \qquad 1.1.1$$

Where:

- m = Pedestrian mass
- τ = Relaxation time (the reaction delay)
- v_0 = Desired walking speed
- $\hat{e}_i$ = The desired velocity vector (speed × direction toward target)
- v_i = The agent's current instantaneous velocity

Logic: This is a correction force. Based on the mean-reverting logic identified by Coffey et al. (1996), agents attempt to reach a desired velocity (v_i^0) within a relaxation time (τ). If an agent is moving slower than their target speed or in the wrong direction, the equation applies a push to close the gap. The target vector ($\hat{e}_i$) is zone-based:

- Funnel Zone ($x < 0$): target vector ($\hat{e}_i$) converges on the midpoint of the vomitory throat.
- Corridor Zone ($x \geq 0$): target collapses to a pure rightward vector, ensuring agents maintain a forward trajectory along the exit corridor.

Social Repulsion ($F_{i,soc}$)

To manage collision avoidance and energy dissipation in the crowd, a pairwise anisotropic repulsion force is implemented:

$$F_{i,soc} = \sum_{j \neq i} w_{ij} \left[k_{soc} \cdot max\left(0, d_{buffer\ soc} - r_{ij}\right) \hat{n}_{ij} - c_{soc} \cdot 1_{[r_{ij} > d_{buffer}]} \left(v_i - v_j\right) \right] \qquad 1.1.2$$

Where:

- w_{ij} = Anisotropy Weight – neighbours in front exert full influence while those behind have reduced impact

$$w_{ij} = \begin{cases} 1.0, & if\ cos\phi_{ij} \geq 0\ (neighbour\ is\ ahead\ or\ beside) \\ \lambda = 0.1, & if\ cos\phi_{ij} < 0\ (neighbour\ is\ behind) \end{cases}$$

- $cos\phi_{ij}$ = Dot product $(-\hat{n}_{ij} \cdot \hat{v}_i)$ which calculates the angle between the agent's velocity unit vector and the direction of the neighbour
- k_{soc} = Repulsion stiffness – how hard the spring pushes back
- $d_{buffer\ soc}$ = Personal space radius – trigger distance of force
- r_{ij} = Straight-line distance between agent i and neighbour j
- $\hat{n}_{ij}$ = Unit repulsion vector – pointing directly away from the neighbour
- c_{soc} = Damping coefficient – resists relative velocity on contact
- $1_{[\cdot]}$ = Indicator Function – ensures damping forces only activate if agents are overlapping

Logic: The first half of the equation (the spring) prevents people from walking through each other. The second half (the damper) is vital for clogging realism; it acts like friction, draining energy when agents collide and allowing the arches to freeze in place rather than vibrating unnaturally.

Boundary Repulsion ($F_{i,bound}$)
This force accounts for the physical stadium walls. It is designed to ensure agents slide along boundaries rather than becoming trapped by perpendicular force spikes:

$$F_{i,bound} = max\left(0, k_{wall} \cdot max\left(0, d_{buffer\ wall} - \delta\right) - c_{wall} \cdot (v_i \cdot \hat{n})\right)\hat{n} \qquad 1.1.3$$

Where:

- k_{wall} = Repulsion stiffness – how hard the spring pushes back
- $d_{buffer\ wall}$ = Wall activation radius – trigger distance of force
- δ = signed distance from the agent to the top or bottom wall
- c_{wall} = Damping coefficient – resists relative velocity on contact
- $\hat{n}$ = Wall normal vector – perpendicular to the wall surface

Logic: In the funnel, the walls are angled, so the normals ($\hat{n}$) are rotated to match the slope. This forces agents to steer into the bottleneck. The outer $max(0,\cdot)$ is a critical limiter; it ensures the wall only ever pushes, never pulls. Without this, an agent moving away from a wall could be sucked back toward it by the damping term – a non-physical failure state identified during the calibration phase as the "Sticky Wall" phenomenon (Section 4.2.2).

Numerical Stability and Calibration Logic
To ensure the simulation remains physically grounded – specifically during the high-pressure stochastic triggers analysed in Section 4 – the following architectural constraints are applied within Component 3. These constraints are based on the stability envelope identified during the heuristic calibration phase (Section 4.2), which resolved the numerical and behavioural anomalies encountered during early-stage testing.

Heuristic Force Weighting: A hierarchy of force magnitudes was established to prioritise architectural and social safety over navigational speed. As detailed in the force hierarchy explanation in Section 4.2.1, the engine scales interaction terms such that Boundary > Social > Driving. This ensures that navigational intent does not neutralise social or architectural constraints, preventing both clipping through boundaries and the unrealistic robotic flow that occurs when agents ignore social spacing to reach their target velocity.

Magnitude Caps: Total resultant forces and velocities are subjected to hard caps. This acts as a circuit breaker for the Euler-Maruyama solver, preventing the force-spikes and instantaneous accelerations that cause agents to non-physically teleport across the geometry in a single time step.

Heuristic Calibration: Beyond global caps, Component 3 incorporates the fine-tuning of specific parameters within the F_{drive}, F_{soc} and F_{bound} equations. This iterative adjustment ensures that the

mathematical interaction logic produces realistic navigation and avoids non-physical artifacts. These parameters were calibrated to the specific thresholds identified in Section 4.2.2.

Component 2: Agent Initialisation and Temporal Funnelling
The initialisation of the crowd is a critical factor in model stability. To prevent the instantaneous force peaks caused by spawning agents in a high-density block, the framework utilises temporal funnelling. As identified in the Corroboration Threshold analysis (Section 4.2.3), these parameters must be precisely synchronised with the corridor geometry to prevent immediate numerical divergence.

Staggered Entry Delay: Agents are released at discrete intervals. Behaviourally, this mirrors the "Pulse-Loading" managed by stadium stewards – a strategy of spreading the initial density over time to prevent numerical instability at the corridor entrance. The calibration of this rate is essential to prevent the permanent clog states identified in the Spawn Rate findings of Section 4.2.3.

Uniform Spatial Distribution: To prevent force-locking (where agents in perfect columns exert symmetrical pressures that render them immobile), agents are distributed across the entrance width using fixed vertical intervals. This ensures a realistic spread while maintaining the numerical convergence of the solver – a prerequisite for stability discovered during the Spawn Position trials detailed in Section 4.2.3.

The Activation Mask: The simulation utilises a Boolean mask to gate the agent array. This ensures that only live agents within the stadium are included in the force summation, isolating the active crowd from ghost forces exerted by agents who have exited or yet to spawn. The mathematical necessity of this mask is discussed as Interaction Gating in Section 4.2.3.

By implementing these structural constraints, the model graduates from a theoretical study into a behaviourally informed stadium egress simulation. The logic is now set for Section 3.3.3, which details the observation protocol used to harvest the results presented in Chapter 4.

3.3.3 Data Extraction and Observation Protocol
To ensure the simulation produces more than visual output, a multi-layered logging architecture was implemented. This allows every micro-state of the crowd to be recorded for retrospective analysis, ensuring that emergent behaviours can be verified mathematically alongside qualitative observation.

The Logging Architecture
The model utilises a Three-Track History system to capture the evolution of the crowd at every time step ($\Delta t = 0.02\text{s}$). This structure ensures that once the physics engine concludes, the following arrays are available for post-processing:

- **Spatial History:** Captures the $[x, y]$ coordinates of every agent. This serves as the primary dataset for generating trajectory plots and verifying boundary adherence.
- **Velocity History:** Logs the instantaneous velocity vectors. This is essential for calculating speed-decay as density increases and identifying the onset of stalling.
- **Density History:** Tracks the real-time occupancy of the bottleneck zone.

Defining Egress Success: Total Egress Time (TET)
The primary quantitative metric for evaluating model efficiency is Total Egress Time (TET). To calculate this, a specific exit threshold was established at $x = 12.0\text{m}$, representing the point where an agent has successfully cleared the stadium vomitory.

An agent is logged as evacuated the moment their x-coordinate exceeds this value. The TET for a specific scenario is defined as the time-step when the final agent in the simulation array crosses this threshold. To account for scenarios where high levels of noise cause total system failure, a flow collapse timeout of 300 seconds (15,000 steps) was implemented. If the final agent fails to exit within this window, the trial is recorded as a mechanical clog or failure, providing a qualitative boundary for the simulation's operational limits.

Bottleneck Analytics: The "Danger Zone"

While the simulation monitors the entire corridor, diagnostic analysis is focused on the "Danger Zone"; the area immediately preceding the vomitory throat. This is defined spatially as the 1.0m approach area where x falls between -1.0 and 0.0.

This region represents the zone of highest lateral pressure. It is here that social repulsion (F_{soc}) and boundary constraints (F_{bound}) are most likely to reach a state of equilibrium that cancels out the forward driving intent (F_{drive}), triggering the formation of an arch. By isolating this zone, the model calculates localised density spikes that would otherwise be averaged out in a whole-corridor analysis.

Visual Diagnostics: Speed-Based Monitoring
To facilitate the qualitative observation of clogs, a speed-based monitoring system was integrated into Component 5. This translates the abstract velocity vectors into a real-time diagnostic map and can be seen in Figure 4.4.

Table 3.3.3: Agent Behavioural State Classification and Velocity Thresholds

Colour	Behavioural State	Velocity Threshold (v)
Green	Laminar / Free Flow	$v \geq 0.8$ m/s
Yellow	Congested / Interaction-Dominated	0.2 m/s $\leq v < 0.8$ m/s
Red	Jammed / Structural Arch	$v < 0.2$ m/s
Black	Inactive (Pre-spawn or Exited)	$x \leq -10.0$ *or* $x \geq 12.0$

The Experimental Sweep Protocol
The final pillar of the methodology is the implementation of a stochastic sweep. To isolate the impact of behavioural noise, the simulation was executed through a batch loop where the Diffusion Coefficient (D) was varied incrementally.

Based on initial verification, D was swept from a deterministic baseline of 0.0 up to a maximum of 2.0 (identified in Section 4.3.3 as the boundary of behavioural realism where trajectories become physically inconsistent) in increments of 0.1. Each D-value was repeated for seven trials to account for the inherent randomness of the Langevin kick, with results reported as a mean TET $\pm$ standard deviation. During these trials, all other heuristic constants remained frozen. This rigorous isolation ensures that any observed fluctuations in efficiency can be mathematically attributed to the stochastic trigger D rather than changes in the physical environment.

This protocol provided the data required to generate the non-linear efficiency curves analysed in Section 4.3, establishing a clear Stochastic Penalty for stadium egress.

4. Results

4.1 Verification: The Stochastic Navigational Engine

The objective of this section is to verify that the mathematical framework defined in Section 3.1 was successfully translated into the Python environment. Verification confirms that the code produces the expected behaviour under controlled conditions before complex scenarios are introduced.

4.1.1 The Deterministic Baseline

The engine was first tested with the Diffusion Coefficient (D) set to zero. Under these conditions, agents followed perfectly linear trajectories toward the target line. As illustrated by the representative samples in Figure 4.1.1, agents followed perfectly linear trajectories toward the target line, confirming that the Social Force Model (SFM) driving terms were correctly integrated and that the Euler-Maruyama solver maintained deterministic stability in the absence of noise.

4.1.2 Stochastic Path Deviation

Following the deterministic check, the diffusion coefficient was increased to observe the impact of navigational uncertainty. As illustrated in Figure 4.1.2, the introduction of the noise variable resulted in the expected jagged trajectories characteristic of Active Brownian motion. While this plot tracks three

representative agents from a population of $N = 50$, quantitative analysis confirmed that the lateral variance of the entire crowd scaled in proportion to the noise intensity.

This provides the mathematical proof that the D variable functions as an effective measure of behavioural agitation. While this implementation utilises a global constant for D to isolate its impact on egress efficiency, the potential for heterogeneous or agent-specific noise profiles is addressed as a further developmental step in Section 5.4.

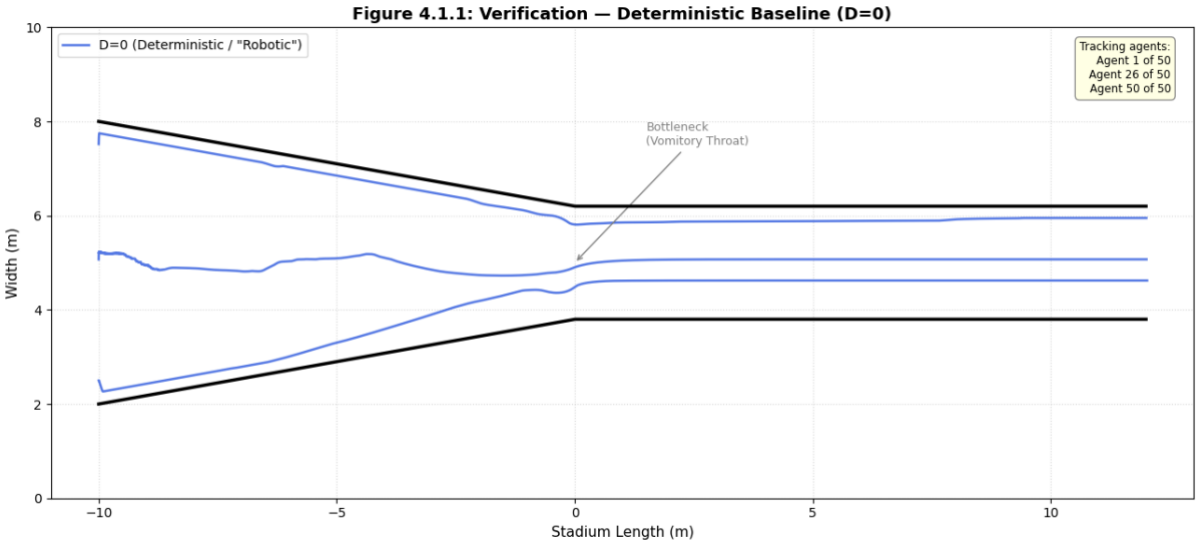


Figure 4.1.1: Trajectory verification under the deterministic baseline (D=0.0). *Representative longitudinal paths for agents 1, 26, and 50 are shown as they navigate the funnel geometry toward the vomitory throat (x=0 m) and along the exit corridor. Solid blue lines represent the path of the agents in the absence of stochastic noise.*

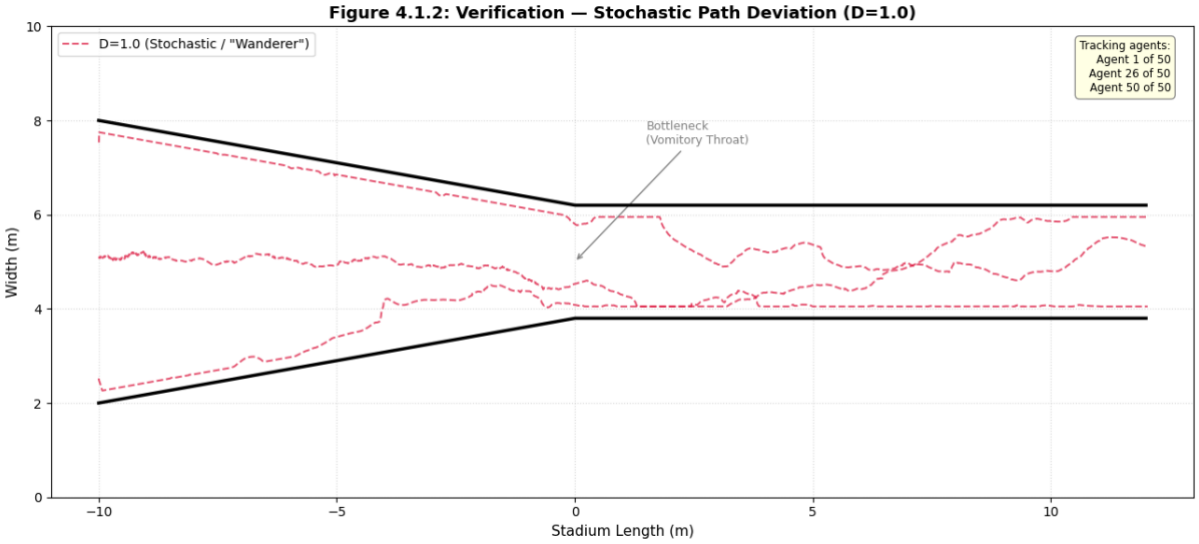


Figure 4.1.2: Trajectory verification under moderate stochastic noise (D=1.0). *Representative longitudinal paths for agents 1, 26 and 50 navigating the vomitory geometry with the Diffusion Coefficient set to 1.0. Red dashed lines represent the lateral deviations from the goal-seeking target vector.*

4.2 Heuristic Calibration: Establishing the Stability Regime

Once the engine was verified, a series of iterative trials were performed to calibrate the interaction forces. This section documents the finding that the model's physical realism is highly sensitive to both the relative weighting of the deterministic forces and the synchronisation of agent parameters.

4.2.1 The Force Hierarchy Finding

Initial iterations of the simulation demonstrated that without strict numerical limiters, interaction forces could reach magnitudes exceeding 50,000 N (far beyond human physical capabilities) during agent overlaps. This resulted in agents teleporting; the resultant acceleration catapulting agents through vomitory boundaries or across the geometry in a single frame. Furthermore, it was observed that if the driving force (F_{drive}) was too dominant, agents prioritised their target line to the exclusion of all social and architectural constraints. This created a robotic and perfectly laminar flow that effectively suppressed the individual-level stochastic triggers required to initiate the collective arching and clogging phenomena qualitatively analysed in Section 4.4.

To resolve both the numerical instability and the lack of behavioural realism, a two-tiered calibration strategy was established. First, the establishing of a Hierarchy of Force Magnitudes ensured that architectural integrity (F_{bound}) and social spacing (F_{soc}) maintained priority over navigational intent (F_{drive}):

$$F_{bound} > F_{soc} > F_{drive} \quad 3$$

Second, global force and velocity caps were implemented to act as circuit breakers for the Euler-Maruyama solver. By capping each force according to the established hierarchy, and velocity at 3.0 m/s, the model prevented the exponential force-spikes typical of spring-model interactions. This calibration was vital for ensuring that agents slid along vomitory boundaries rather than bouncing uncontrollably or clipping through the geometry, providing the necessary baseline for the high-density bottleneck analysis in Section 5.

4.2.2 Numerical Stability Envelope

The calibration phase identified the break points of the simulation logic, specifically concerning the EM solver's convergence. Table 4.2.2 summarises the final calibrated constants that defined the stability regime. These values represent the numerical configuration required to maintain the Thinking Fluid behaviour without divergence.

Table 4.2.2: Final Calibrated Simulation Constants

Parameter	Unit	Value	Correction Log / Finding
Δt	s	0.02	Balance visual fluidity and solver stability. Values >0.05 s caused stochastic divergence.
m	kg	70	Standard average human mass for realistic inertial calculations.
τ	s	0.5	Low values caused jittery over-correction; high values led to sluggish response.
v_0	m/s	1.34	Establishes the baseline intent of the crowd. Derived from Weidmann (1993) empirical mean.
k_{wall}	N/m	7500	Excessive stiffness without dampening caused agents to stick to boundaries upon contact.
k_{soc}	N/m	5000	Ensures agents deform the crowd shape rather than clipping through neighbours.
$d_{buffer\ wall}$	m	0.25	Defines proximity threshold for wall interactions to prevent geometry penetration.
$d_{buffer\ soc}$	m	1.0	Defines the radius of realistic personal space (Helbing, 1995).
c_{wall}	Ns/m	1000	Critical dampening required to resolve "Sticky Wall" locking and allow smooth sliding.
c_{soc}	Ns/m	750	Dampening term used to mitigate oscillatory bouncing during dense interactions.
F_{drive} Cap	N	1000	Prevents intent from overwhelming social forces and creating robotic, laminar flow.
F_{soc} Cap	N	3500	Mid-tier of the Force Hierarchy; prevents social pressure from inducing solver failure.
F_{bound} Cap	N	5000	Apex of the Force Hierarchy; ensures architectural integrity against high-noise kicks.
λ	-	0.1	Anisotropic vision weighting; prevents unnatural backward movement by rear neighbours.

v_i Cap	m/s	3.0	Prevents non-physical teleportation and keeps movement within biological limits.

To achieve the physical realism required for the bottleneck analysis, several specific behavioural anomalies had to be resolved through iterative testing following the implementation of the force hierarchy.

The "Sticky Wall" Phenomenon (F_{bound}): In later iterations, boundary repulsion lacked sufficient dampening. This resulted in agents locking onto the vomitory walls upon close proximity, unable to break away due to the high repulsive gradient. This observed behaviour validated the necessity of the $max(0,\cdot)$ limiter logic defined in Section 3.3.2. Precise calibration of the wall buffer and stiffness was also required to ensure agents slid along the geometry rather than becoming trapped by it.

Anisotropic Social Perception (λ): The implementation of a weighting constant ($\lambda = 0.1$) for the rear-facing vision cone within F_{soc} was a vital finding. Without this weighting, agents reacted with equal force to pedestrians behind them as those in front. This led to unrealistic behaviour where agents would walk backward or stall unnecessarily. The 0.1 factor provided a realistic blind spot, focusing agent interactions on their forward path.

4.2.3 The Corroboration Threshold: Agent Parameter Integration

A critical finding during calibration was the identification of the Corroboration Threshold. It was observed that the three primary agent initialisation parameters (Spawn Rate, Spawn Position and Exit Logic) cannot be treated as independent variables; they must be synchronised with the corridor geometry to produce a stable crowd scenario.

Spawn Position (Vertical Offsetting)**:** Initial trials where agents were spawned at a singular coordinate resulted in non-physical stacking, where repulsive forces reached infinite peaks and crashed the solver. The implementation of fixed vertical intervals across the entrance width was found to be a prerequisite for stability. This lateral distribution ensured that agents entered the corridor in a realistic spread, preventing force-singularities and allowing for the development of laminar flow. While this provides a stable baseline, the integration of truly random lateral jitter is identified as a further step for behavioural realism in Section 5.4.

Spawn Rate (vs. Geometry)**:** Observations confirmed that the entry rate must be calibrated against the width of the vomitory throughout. If the spawn rate exceeded the physical capacity of the bottleneck, the system reached a state of permanent clog immediately. In these instances, the Diffusion Coefficient (D) became irrelevant, as the system was overwhelmed by density alone.

Exit Logic Synchronisation: The definition of the exit condition was found to be a primary driver of flow stability. Initial iterations utilising a single-point exit coordinate caused agents to artificially converge, resulting in non-physical circular swirls at the bottleneck. The transition to a Target Line allowed for a more natural longitudinal flow, utilising the full width of the exit.

Interaction Gating: A fundamental requirement for physical realism was the implementation of a gating mechanism to isolate force interactions within the active simulation domain. By utilising a Boolean mask to filter the agent array, the framework successfully eliminated ghost forces; forces caused by agents who had either not yet spawned or had already evacuated the corridor. Restricting the force summation exclusively to live agents was essential for preserving the integrity of the bottleneck dynamics and ensuring that the observed clogging was driven solely by active crowd interactions rather than ghost influences from outside the stadium geometry.

These findings demonstrate that a successful egress simulation is a finely balanced ecosystem. The necessity of this synchronisation is discussed in Section 5.3 as a direct parallel to the real-world management of "Pulse-Loading" in stadium environments.

4.3 Quantitative Analysis: Behavioural Noise (D) vs. Egress Efficiency

The primary objective of this section is to quantify the relationship between the Diffusion Coefficient (D) and the overall efficiency of the egress process. While Section 4.2 established the stability of the engine, this analysis isolates behavioural noise as the independent variable to determine its impact on Total Egress Time (TET) (Figure 4.3), utilising the batch-loop protocol defined in Section 3.3.3.

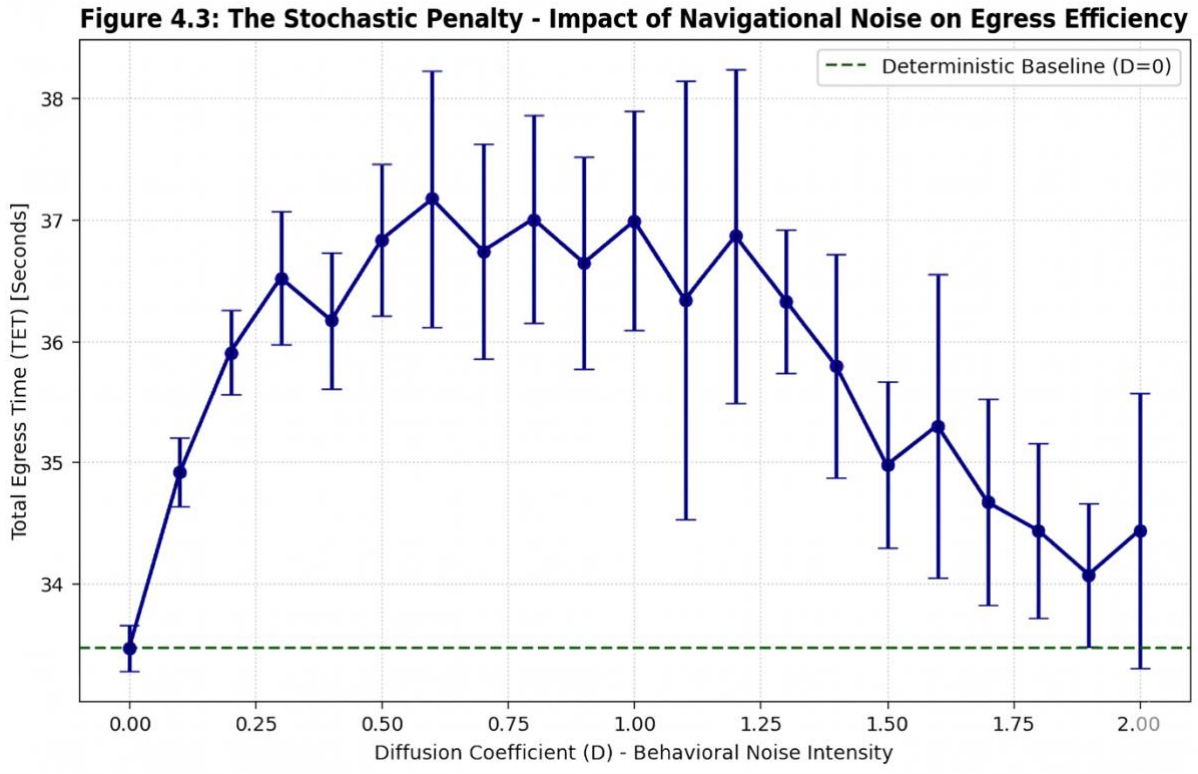


Figure 4.3: Relationship between Total Egress Time (TET) and the Diffusion Coefficient (D)*. The mean TET (blue markers) and standard deviation (error bars) were derived from seven independent simulation trials for each 0.1 increment of D. The horizontal green dashed line represents the deterministic baseline clearing time of 33.4 s achieved at D=0.0.*

4.3.1 Zone 1: The "Stochastic Penalty" Onset ($D = 0.0$ to 0.5)

The results reveal an immediate and statistically significant "Stochastic Penalty" – a non-linear efficiency loss triggered by the introduction of navigational noise. At the Deterministic Baseline ($D = 0.0$), the crowd clears the exit in 33.4 seconds, represented by the green dashed line in Figure 4.3. This represents the theoretical capacity of the geometry under perfect, laminar conditions.

As D increases to 0.5, the TET rises to a peak of approximately 37.4 seconds, representing an 11% efficiency loss. The standard deviations in this range remain tight ($\pm$0.3-0.6s), indicating that the correlation between noise and delay is statistically robust. This confirms that even minor levels of behavioural agitation genuinely degrade egress throughout, directly challenging the mean-flow assumptions of current design standards.

4.3.2 Zone 2: The Saturated Plateau ($D = 0.5$ to 1.25)

Beyond the initial penalty onset, the system enters a saturated plateau. In this regime, the TET oscillates between 35s and 37.5s without exhibiting a further monotonic increase. This plateau suggests that the 2.4m bottleneck has reached a physical failure threshold; the geometry is sufficiently narrow that once noise triggers the onset of clogging, additional agitation cannot worsen the already-degraded throughout significantly.

A key finding in this zone is the widening of the standard deviations ($\pm$0.5-1.05s), visible in Figure 4.3. The overlap in error bars across this range signifies that outcomes are no longer strictly predictable. In some runs,

a stable arch forms quickly, while in others, the stochastic kicks happen to break the clog temporarily. This variance represents a significant finding, as it quantifies the inherent unpredictability of crowd outcomes under moderate noise – a critical risk factor that deterministic models are fundamentally incapable of capturing.

4.3.3 Zone 3: The Dissolution Boundary ($D = 2.0$)

At the upper limit of the investigation ($D = 2.0$), the TET data point in Figure 4.3 exhibits a notable drop back toward the baseline (~34.4 seconds). While numerically interesting, this represents a dissolution boundary. Qualitative verification of the simulation animations (Section 4.4) confirmed that at this magnitude, the navigational noise is so extreme that it disrupts the formation of stable arches faster than it creates congestion.

As these trajectories exhibit jaggedness inconsistent with the biological constraints of human motion, $D = 2.0$ is established as the boundary of behavioural realism for this model. Consequently, data points beyond this threshold were excluded to maintain the integrity of the stadium egress narrative.

4.3.4 Localised Bottleneck Dynamics: The Density-Duration Proof

While the qualitative observations in the preceding sections identify the presence of arches, the density history log provides the mathematical proof of their impact on the bottleneck environment. By analysing the occupancy of the approach zone ($x \in [-4.0, 0.0]$ m), a clear quantitative signature of stochastic failure emerges.

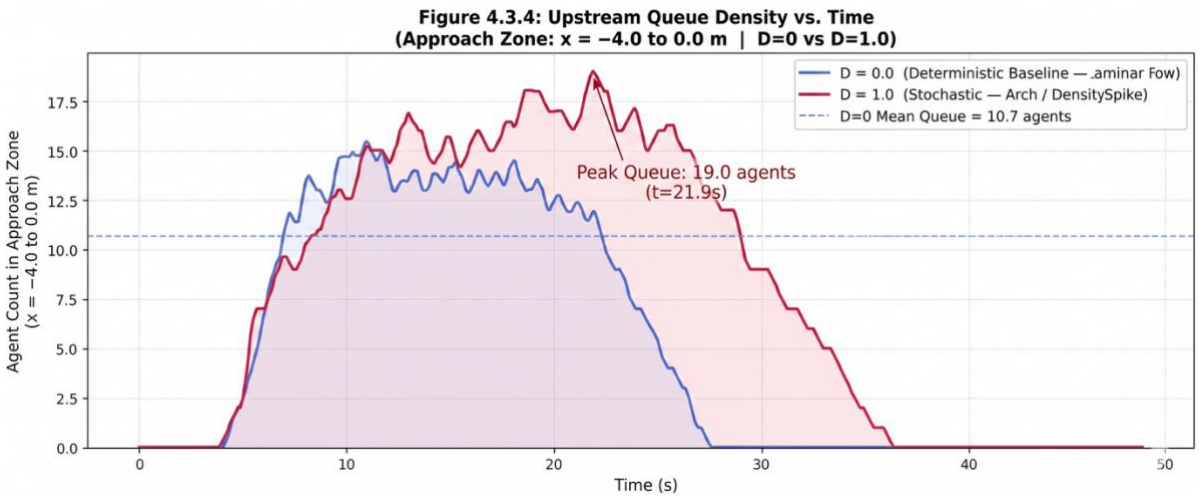


Figure 4.3.4: Upstream queue density over time within the bottleneck approach zone (x=−4.0 to 0.0 m). *The agent count is compared between the deterministic baseline (D=0.0, blue solid line) and a stochastic scenario with moderate behavioural noise (D=1.0, red solid line). The peak queue for the stochastic trial is annotated at 19.0 agents (t=21.9 s). The dashed horizontal line represents the mean queue density for the D=0 trial (10.7 agents).*

As illustrated in Figure 4.3.4, the primary impact of behavioural noise results in a fundamental shift in egress duration alongside an intensified density peak:

- **Deterministic Baseline ($D = 0.0$):** The approach zone reaches a peak occupancy of approximately 15.5 agents at $t \approx 11$ s. The queue clears sharply and symmetrically, with the zone emptying completely by $t \approx 27.5$ s, representing efficient, laminar throughput.
- **Stochastic Analysis ($D = 1.0$):** Under moderate behavioural noise, the peak occupancy rises to 19 agents at $t \approx 21.9$ s. Crucially, the density plot exhibits an extended residual tail, where the zone remains occupied until $t \approx 36.5$ s.

This residual tail represents the intermittent arching identified in Section 4.4, where throughput becomes irregular batches rather than a steady stream. The additional 9 seconds of occupancy directly corroborates the global TET increases in Section 4.3, confirming that global delays are the result of localised stalling events.

Ultimately, this reframes the Stochastic Penalty as a temporal risk: behavioural noise intensifies the peak pressure and prolongs the high-pressure state at the bottleneck. For stadium design, this implies that infrastructure must be resilient to sustained lateral pressures for significantly longer periods – in this study, a 33% increase in duration – than deterministic mean-flow calculations suggest.

4.4 Qualitative Observation: Clogging and Arching

While Section 4.3 quantified the efficiency loss, this section provides a qualitative examination of the physical behaviours that caused those delays. By utilising the velocity-coded visualisation environment (Component 5), it was possible to observe the specific sequence of events that led to the formation of a bottleneck clog.

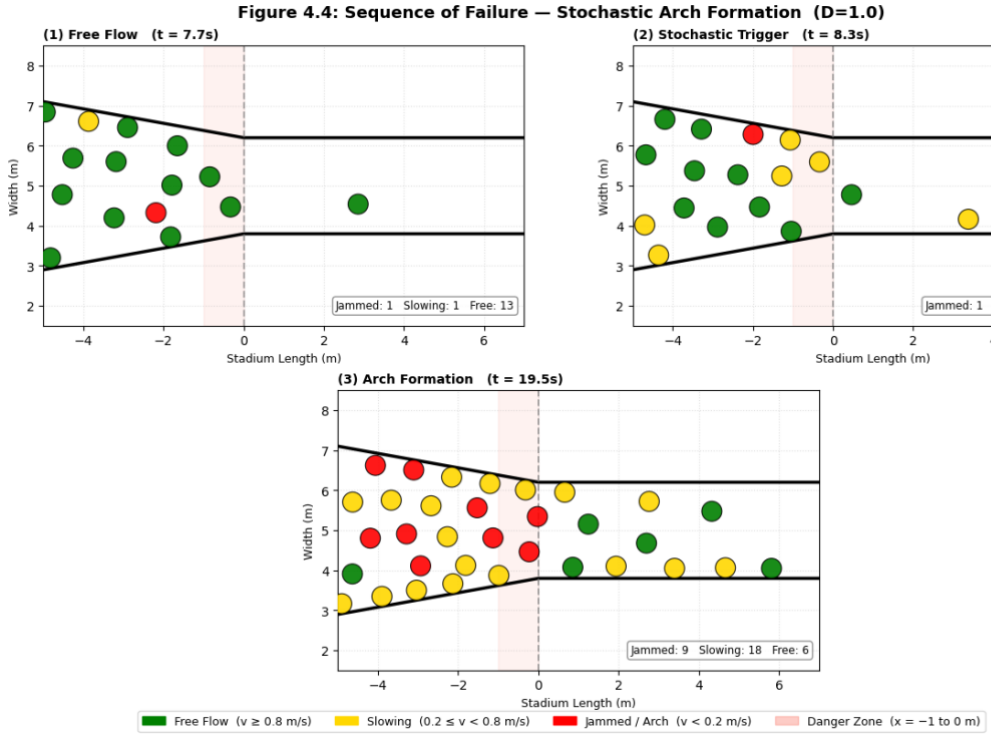


Figure 4.4: Simulation snapshots showing the sequence of failure and stochastic arch formation (D=1.0). A full video animation of this specific trial sequence is available via the GitHub repository link provided in the Abstract. *The sequence illustrates (1) initial laminar flow at t=7.7 s, (2) the stochastic trigger causing a ripple of deceleration at t=8.3 s, and (3) the resulting structural arch formation at t=19.5 s. Agents are color-coded as per* ***Table 3.3.3****. The shaded Danger Zone indicates the area of maximum lateral pressure (x=−1.0 to 0.0 m).*

4.4.1 Visual Observation of Arch Formation

In trials where behavioural noise was introduced ($D \geq 0.1$), the simulation consistently exhibited the emergence of semi-circular arches at the vomitory throat. As seen in the three-stage sequence in Figure 4.4 and quantitatively proven by the prolonged residual density identified in Section 4.3.4, these arches are characterised by a keystone group of agents whose velocity drops to near zero ($v < 0.2$ m indicated in red). By $t = 19.5$ s (Frame 3), the jammed count increases to 9 agents, representing the full maturation of the arch across the entrance of the exit corridor.

This stationary group creates a physical barrier at the centre of the exit. Because trailing agents are governed by the social repulsion force (F_{soc}), they are forced to stall or deviate to avoid a collision, regardless of their individual intent to exit. This confirms that the clogging observed in the data is a physical structural event triggered by the interaction of agents at the bottleneck. This is corroborated by the density history log, which shows localised occupancy in the Danger Zone ($x = -1.0$ to 0.0 m) sustaining elevated levels for approximately 9 seconds beyond the deterministic baseline during these arching events (see Section 4.3.4).

4.4.2 Observation of Cascading Deceleration

A key observation during the high-noise animations was the sequence of deceleration. As illustrated in frame (2) of Figure 4.4 at $t = 8.3$ s, systemic stalling typically began with a single agent experiencing a high lateral kick from the D variable at the bottleneck throat.

Once an individual agent hesitated or shifted laterally at the entrance of the 2.4m corridor, a ripple of deceleration (yellow slowing states) was observed to propagate backward through the queue. This sequence, where a minor individual perturbation leads to a collective stall, suggests a highly sensitive interaction mechanism. The specific theoretical mechanics of this ripple effect are examined in greater detail in Section 5.2.

4.4.3 Interaction Integrity: Agitation vs. External Force

Critically, these clogging events were observed to occur even when the external driving force (F_{drive}) remained constant. In every trial, the agents maintained a consistent mathematical intent to move toward the exit line.

This observation indicates that the transition from flow to failure within this model is an internalised process driven by agent agitation (D). Even as the keystone agents in frame (3) of Figure 4.4 reach a near-stationary state, their internal intent to reach the target remains constant.

The simulation demonstrates that the system can reach a point of failure without an increase in rearward crowd pressure, providing a visual counter-narrative to traditional hydraulic crowd models. The implications of this internal agitation for stadium design are discussed in Section 5.2.3.

5. Discussion

5.1 Critiquing the Mean-Flow Assumption

As established in the appraisal of current industry standards (Section 2.1.3), benchmarks such as The Green Guide (Sports Grounds Safety Authority, 2018) largely rely on macroscopic, deterministic assumptions to define safe egress. These standards utilise a mean-flow approach, where the capacity of an exit is calculated based on a linear relationship between corridor width and the number of pedestrians passing through per minute. However, the results observed in this research, specifically the transition from laminar flow to stochastic clogging, suggest that these deterministic calculations lack the necessary resilience to account for behavioural variance. Consequently, this research addresses a critical oversight in current safety modelling by quantifying how individual-level uncertainty physically reduces the effective capacity of an exit route.

5.1.1 The "Deterministic Fallacy": Humans as Robotic Agents

The mean-flow assumption essentially treats every spectator as a perfect mean-velocity follower, equivalent to the $D = 0$ verification test performed in Section 4.1.1. In such a system, flow is perfectly predictable and safety is a simple function of geometry. However, as demonstrated in Section 4.3, once navigational noise (D) is introduced, the relationship between corridor width and efficiency becomes non-linear.

The research shows that even at densities considered safe by traditional standards, a high degree of navigational uncertainty can act as a catalyst for a systemic flow collapse, rendering the deterministic prediction inaccurate. This is evidenced by the 11% efficiency loss observed as D increased to a moderate 0.5 in Section 4.3.1, a threshold where The Green Guide (2018) baseline fails to anticipate the onset of delay. This finding aligns with the principle that diversity congests, established by Campanella et al. (2009), who identified that speed variance alone can significantly degrade throughput in bottlenecks, even before physical contact forces dominate.

5.1.2 Dynamic Safety Margins vs. Static Widths

The primary implication of these findings is that a one-size-fits-all approach to corridor width is insufficient for modern stadium safety. If a crowd is agitated, whether due to high-stakes rivalry, alcohol consumption, or high-urgency levels, the effective width of the vomitory is effectively reduced by the stochastic oscillations of the agents. This suggests that safety margins should be dynamic rather than static, scaling according to the expected behavioural profile of the event:

- **Low-Noise Events:** (e.g., family-oriented matinees) may align closely with current Green Guide benchmarks.

- **High-Noise Events:** (e.g., evening matches with identified behavioural stressors) require a stochastic buffer. This model indicates that in these scenarios, the navigational noise creates a physical interference that mimics a narrower corridor, requiring an increase in designed exit capacity to maintain the same level of safety.

5.1.3 The Stochastic Resilience of the Langevin Approach

By moving away from mean-flow and adopting an Active-Langevin perspective, engineers can identify the specific point of failure for a unique demographic. Rather than assuming a constant flow rate, the results in Section 4.3.1 demonstrate the necessity of designing for worst-case noise; at $D = 0.5$, the model recorded a peak efficiency loss of 11%, identifying the specific level of agitation where the deterministic throughput first collapses.

Furthermore, the high variance of outcomes observed in the plateau regime (Section 4.3.2) and the prolonged residual occupancy recorded in the Danger Zone (Section 4.3.4) prove that as noise increases, the system becomes governed by the temporal duration interaction-delays rather than purely geometric capacity. This provides a more resilient safety framework, as it accounts for the individual-level hesitations and erratic movements that act as the actual triggers for systemic flow failure. This approach allows for the development of safety standards that are not only architecturally sound but behaviourally informed.

5.2 The Mechanics of the Stochastic Trigger

The transition from a free-flowing laminar state to a clogged state is often treated in civil engineering as a function of density alone. However, the results in Section 4.4 demonstrate that density is merely a prerequisite for failure; the actual catalyst is stochastic interaction. While a deterministic model treats a crowd as a continuous fluid (similar to water flowing through a pipe) the Active-Langevin engine reveals that pedestrians behave as discrete thinking particles whose internal fluctuations can override global flow intent.

5.2.1 Individual Hesitation as a Physical Force

Within the simulation framework, the stochastic noise term (ξ) manifests physically as a sudden hesitation, a lateral shift, or a momentary deviation from the optimal path. While these stochastic kicks are negligible in low-density environments, they become physically coupled through the social repulsion force (F_{soc}) as the crowd approaches the bottleneck. This coupling transforms individual navigational uncertainty into a collective structural phenomenon.

When an agent experiencing a high kick from the Diffusion Coefficient (D) hesitates at the vomitory throat, they do not merely slow down; they create a stationary obstacle. This sequence is visually demonstrated in Figure 4.4, where an individual stochastic trigger at $t = 8.3$ s (Stage 2) initiates a ripple of deceleration through the queue, culminating in the structural arch observed at $t = 19.5$ s (Stage 3). Because the agents behind are governed by the same F_{soc} and F_{bound} constraints, they must decelerate or deviate to avoid a collision. This creates a "Butterfly Effect" where a single random perturbation at the front of the queue triggers a non-linear ripple of deceleration throughout the entire system.

Critically, the density history log (Figure 4.3.4) reveals that this ripple does not merely slow the crowd but compresses it, resulting in a higher peak occupancy than the deterministic baseline. This suggests that stochasticity acts as a physical catalyst, momentarily increasing localised density to levels that trigger the formation of stable, self-reinforcing arches. Consequently, agents are held in the Danger Zone in a state of residual congestion, physically extending the high-pressure environment for nine seconds beyond the deterministic expectation.

5.2.2 Validating the "Thinking Fluid" Hypothesis

This observation provides strong validation for the "Thinking Fluid" hypothesis (Hughes, 2003). Unlike traditional fluid dynamics, where a particle is carried passively by system pressure, a thinking fluid is comprised of discrete entities capable of individual decision-making. In contrast, the results in Section 4.3 show that as D increases, the probability of an arch forming increases, even when the driving force (F_{drive}) remains constant.

This suggests that arching is a structural phenomenon triggered by behavioural uncertainty rather than external pressure. This offers a specific microscopic explanation for the macroscopic crowd turbulence observed by Helbing et al. (2007); whereas the results identify phase transitions in real-world disaster data, this Langevin framework reproduces the specific D kicks that serve as the initiation mechanism for the collapse.

5.2.3 Re-evaluating the Failure Mechanism: Agitation over Pressure

A significant finding of this analysis is that the transition from flow to failure is driven primarily by the internal agitation of the agents rather than external system pressure alone. By maintaining constant values for the driving force (F_{drive}), boundary repulsion (F_{bound}), and corridor geometry while exclusively varying the Diffusion Coefficient (D), this research has successfully isolated behavioural uncertainty as the primary independent variable for failure.

This causal link is established through the comparison of the verification trials (Figures 4.1.1 and 4.1.2), which demonstrate that under identical physical constraints, the presence of navigational noise is the sole catalyst for path deviation. Furthermore, the resulting Stochastic Penalty curve (Figure 4.3) provides the overarching proof that Total Egress Time clearing is dictated by internal agitation levels rather than architectural capacity alone.

The data confirms that a crowd can self-clog through internal hesitation and social repulsion before any external pressure from the rear of the crowd is applied. This fundamentally shifts the safety perspective: the point of failure in a stadium vomitory is as much a function of the crowd's psychological state (D) as it is the physical number of people (N) or the structural width of the corridor. This is visually evidenced by the sequence of failure (Figure 4.4), where the transition to a structural arch is initiated by a single agent's internal hesitation rather than a rearward density surge. This is also quantitatively reinforced by the findings in Section 4.3.4 (Figure 4.3.4), which demonstrate that systemic failure is triggered internally; the bottleneck remains occupied longer not because of more people, but because agitation has compromised the efficiency of the Thinking Fluid.

5.2.4 The Structural Significance of Individual Variance

This shift in perspective; from external pressure to internal trigger, carries profound implications for structural resilience. If a single hesitation can immobilise a crowd, the engineering priority must move from static width to absorbing these stochastic shocks over time. The risk identified in Section 4.3.4 is temporal as well as physical: infrastructure must be designed to withstand design-load densities for significantly longer periods – in this study, a 33% increase in duration – due to these stochastic delays. Consequently, accounting for individual-level variance is a fundamental requirement for predicting the true, time-dependent stress placed on high-stakes egress routes.

5.3 System Implications: Interdependence and Stewardship

The transition from a stable egress to a systemic collapse is rarely the result of a single isolated variable. Instead, the findings in Chapter 4 indicate that the resilience of a stadium vomitory is dictated by the precise calibration of interdependent parameters.

5.3.1 The Corroboration Threshold

As explored in the calibration phase (Section 4.2.3), this research identifies a critical corroboration threshold; a state where the simulation's internal parameters must align with the physical geometry to produce valid results. The model functions not as a collection of isolated forces, but as a balanced ecosystem where spawn rate, initial spatial distribution, and exit logic are inextricably linked to the chosen vomitory width. Observations revealed that if any one of these parameters was unsynchronised (for example, if the spawn rate exceeded the physical capacity of the corridor) the system entered a state of permanent clog where the bottleneck failed instantly regardless of individual agent behaviour.

In these states of oversaturation, the behavioural noise (D) becomes secondary as the system is overwhelmed by pure density. This proves that stochastic safety analysis only provides meaningful insights within a specific window of operational stability. Ultimately, if the basic physical logistics of the crowd and the

architectural limits are poorly managed, no amount of individual-level behavioural modelling can prevent a systemic failure.

5.3.2 Engineering Proxy: "Pulse-Loading" and Infrastructure

While the implementation of a staggered spawn rate was initially established as a numerical necessity to prevent force-singularities and solver divergence (as detailed in Section 3.3.2), the results in Section 4.2.3 reveal it to be a primary finding for egress resilience. In this context, the spawn rate variable serves as a direct mathematical proxy for the arrival rate of a crowd, which is physically controlled by either stewardship or gate infrastructure.

While researchers like Chu (2019) often frame stewards primarily as informational guides who provide layout familiarity, these results suggest a more critical engineering role as temporal managers of crowd density. This leads to the significant recommendation that entry management should be viewed as a "Pulse-Loading" strategy: the deliberate, staggered synchronisation of spectator arrival rates with the known physical capacity of the exit infrastructure.

Just as the simulation requires a controlled entry rate to prevent a non-physical force peak, stadium safety requires the synchronisation of spectator entry with the known physical capacity of the exit routes. This model demonstrates that if the entry rate is not synchronised with the infrastructure's width, a disaster-level density spike is inevitable, regardless of how well-behaved or calm the individual agents may be. This shifts the engineering focus from purely static corridor widths to the active, temporal management of crowd arrival rates.

5.3.3 The Active-Langevin Contribution

The overarching contribution of this research is the introduction of a framework that treats a crowd as a goal-oriented but stochastic medium. Traditional deterministic models often view behavioural variance as a mathematical error or an outlier to be averaged out, whereas the Active-Langevin approach forces the engineer to consider individual-level uncertainty as a core design constraint.

By moving away from robotic mean-flow assumptions, this framework provides a more resilient safety platform. It acknowledges that the worst-case scenario in a stadium is not merely a high-density crowd, but an agitated one. Designing for the Stochastic Penalty ensures that infrastructure is built to absorb the erratic, non-linear hesitations that characterise human movement, providing a safety margin that is behaviourally informed rather than just architecturally sufficient.

5.4 Critical Evaluation and Future Research

The Active-Langevin framework established in this study demonstrates the significant impact of behavioural noise on egress efficiency. However, several simplifying assumptions were necessary to isolate the Stochastic Penalty. The following sections evaluate these limitations and provide a roadmap for the next generation of behaviourally informed safety models.

5.4.1 Navigational Realism: From Fixed to Stochastic Spawning

The current initialisation strategy utilises fixed vertical intervals to distribute agents, a necessary compromise to ensure numerical stability. Early testing indicated that purely random spawning frequently resulted in force-singularities, where agents overlapping at $t = 0$ generated near-infinite repulsive forces that caused the solver to diverge.

While fixed intervals successfully established a spread crowd, they represent a high degree of order rarely present in a messy real-world stadium concourse. Future research should implement random lateral jitter integrated with a collision-check spawning algorithm. This would allow for the integration of the anticipatory avoidance models pioneered by Xu et al. (2014) and the implicit coordination logic of Karamouzas et al. (2017). Combining these intelligent navigation layers with the stochastic noise of this study would test whether proactive coordination serves to dampen or accelerate the formation of the force-locks observed in Section 4.4.

5.4.2 Demographic and Geometric Complexities

To maintain clarity in the force-interaction analysis, this model utilised a homogeneous agent profile with a mass of 70kg and a uniform relaxation time (τ). However, a real stadium demographic is significantly more diverse, comprising families, children, and elderly spectators.

Future iterations should introduce a distribution of masses and τ values, utilising the heterogeneity framework established by Campanella et al. (2009). A slower relaxation time in a portion of the crowd may exacerbate the Butterfly Effect identified in this study, as these agents would take longer to recover from a stochastic kick, potentially creating more persistent obstacles at the bottleneck.

Furthermore, moving beyond the rectilinear vomitory used to isolate bottleneck physics, future work should investigate how corner-cutting behaviours in angled geometries (as found in real-world stadium exits) interact with D to alter the Stochastic Penalty curve.

5.4.3 Integrating Cognitive Noise and Wayfinding

The present model treats noise as a purely physical-navigational variable. A vital future development involves integrating navigational familiarity, addressing the cognitive gap identified by Chu (2019). This would allow the model to account for how an agent's knowledge of the venue influences their path selection. As noted in the research, unfamiliarity with a layout often leads to sub-optimal route choices and increased hesitation. Within an Active-Langevin framework, this lack of route knowledge would manifest as a secondary layer of cognitive noise, potentially triggering the same systemic delays as the physical agitation modelled in this study.

By incorporating the perception-based noise models of Iyer et al. (2024), where agents navigate based on specific 3D vision angles, the framework could account for how occlusion (not being able to see the exit) increases an agent's internal agitation. This would transform D from a global constant into a dynamic, density-dependent variable. A dynamic D model would simulate the feedback loop of anxiety seen in high-density crushes, providing a more sophisticated look at the systemic point of failure.

5.4.4 Empirical Calibration of the Stochastic Variable (D)

Finally, the numerical values of D utilised in this study must be calibrated against empirical data to move from a theoretical volume knob to a predictive engineering tool. While established behavioural research – such as the speed-variance studies by Gales et al. (2025) and the motivation-based findings of Young et al. (2021) – have already identified how factors like alcohol consumption or high-stakes rivalry increase crowd agitation, a primary future step involves mapping these physical states to specific numerical values of D. By formalising this link, safety engineers could develop stochastic safety margins specifically tailored to the unique behavioural profile of a given event, ensuring that architectural designs are resilient to the precise levels of noise expected in real-world scenarios.

6. Conclusion

6.1 Synthesis: Answering the Research Questions

Addressing Research Question 1: Computational Feasibility and Stability

- The study successfully proved that it is computationally feasible to integrate individual-level behavioural noise within a stable, agent-based framework.
- Verification trials (Section 4.1) confirmed that the Social Force Model (SFM) driving terms were correctly integrated with the Euler-Maruyama solver, maintaining numerical stability in the absence of noise.
- Calibration of the force hierarchy ($F_{bound} > F_{soc} > F_{drive}$) and the implementation of magnitude caps acted as a vital circuit breaker for the solver, preventing non-physical accelerations during high-density interaction.
- The identification of a corroboration threshold for initialisation parameters ensured that spawn rates and positions were precisely synchronised with the bottleneck geometry, preventing the force-singularities that would otherwise crash the solver.

- Consequently, this project fulfils RQ1 by establishing a robust architecture where the macroscopic intent of the crowd is preserved even under high-noise conditions.

Addressing Research Question 2: The Impact of Individual Randomness on Flow Failure

- The investigation into RQ2 quantified a significant and non-linear Stochastic Penalty triggered by behavioural agitation.
- At a moderate noise level of $D = 0.5$, the model recorded a peak efficiency loss of 11% compared to the deterministic baseline, representing the point where laminar flow collapses into intermittent, arch-driven stalling.
- Visual analysis (Section 4.4) confirmed this failure is mechanistically driven by individual stochastic kicks that propagate as ripples of deceleration, forming structural arches without any increase in rearward crowd pressure.
- Critically, localised analysis revealed that noise compounds risk by simultaneously increasing peak occupancy of the bottleneck and extending the high-pressure duration by 9 seconds – a 33% increase in time-dependent stress.
- Together, these findings confirm that infrastructure must be resilient to both higher peak loads and sustained loading for significantly longer than mean-flow deterministic calculations predict.

6.2 The Case for Stochastic Safety Margins

A primary conclusion of this research is that traditional safety benchmarks, such as The Green Guide, are insufficient for managing the behavioural variance of real-world crowds. Current industry standards utilise static, mean-flow calculations that assume a linear relationship between corridor width and egress capacity.

However, this study demonstrates that once navigational noise is introduced, the relationship between geometry and efficiency becomes non-linear. Navigational uncertainty acts as a physical catalyst for systemic failure, requiring the engineering focus to shift from static widths to dynamic safety margins. Infrastructure must be designed to absorb these stochastic shocks over time, accounting for the fact that agitated crowds stay dangerous for longer.

6.3 Beyond the Stadium: Universal Bottleneck Resilience

While this study utilised a 2.4m stadium vomitory as its primary testing ground, the Active-Langevin framework developed here has broad applications across the built environment. The methodology provides a universal tool for assessing bottleneck resilience in several high-stakes contexts:

- **Mass Transit Infrastructure:** Evaluating Pulse-Loading strategies in metro stations to prevent density-driven force-locks.
- **Emergency Evacuations:** Modelling the increased behavioural noise characteristic of high-urgency or unfamiliar environments to set more realistic clearance timelines.
- **Urban Planning:** Designing walkways and transport hubs that are behaviourally informed, ensuring that architectural boundaries can withstand sustained lateral pressures for significantly longer periods than mean-flow models suggest.

The success of this framework in parameterising the Diffusion Coefficient (D) opens the door for future empirical calibration against real-world data, such as alcohol consumption or demographic diversity, to create event-specific safety profiles.

6.4 Final Summary

This research demonstrates that stadium safety is not merely a function of architectural width, but of both temporal and structural resilience. By proving that individual-level agitation increases the peak intensity of the load and prolongs the duration of bottleneck danger, this project provides a new stochastic lens for engineering a safer built environment. Ultimately, the transition from flow to failure is a behaviourally triggered event; by accounting for the Stochastic Penalty of behavioural noise, civil engineers can establish more resilient safety margins to absorb both the sustained loading and the elevated pressures stemming from the inherent uncertainties of human movement.

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